

Satellite cloud data over France

What exists, what it shows,
and what our Thermal-planes study needs

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Data survey, 7 October 2026

Geostationary

A film of the whole day

An image every 2.5–15 min

Pixels of 0.6–5 km

Polar orbiting

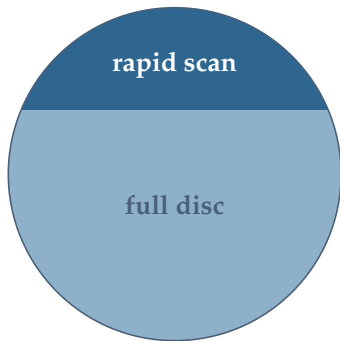
A sharp photo, once or twice a day

Pixels of 10 m–1 km

Sentinel-2, Landsat, MODIS, VIIRS

Geostationary images follow the clouds through the day. A polar image, on the few days it exists, shows the single cumuli that the coarser pixels merge.

Full disc or rapid scan: the whole view or only Europe



the Earth as the satellite sees it

Full disc

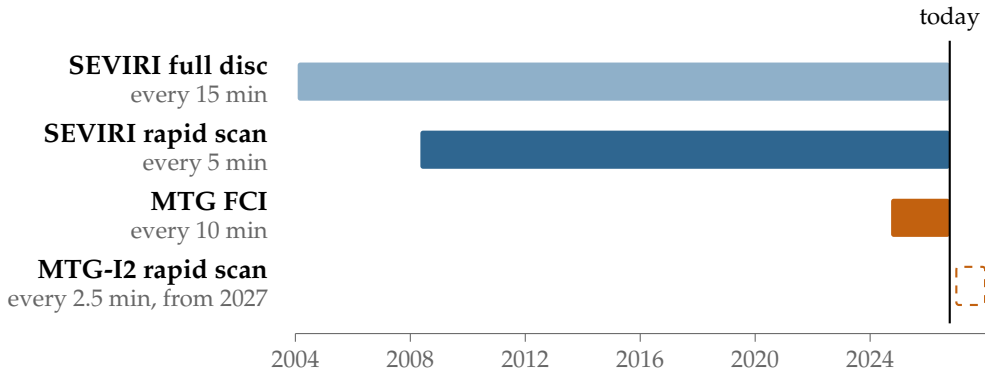
All the Earth the satellite sees, about 40% of the globe

Rapid scan

Only the northern third: Europe and North Africa. Smaller area, scanned more often: with SEVIRI every 5 min instead of 15.

France is in both

Which geostationary satellite covers which years



Three kinds of channels, three questions

Visible

daytime

Where are the clouds?

Near-infrared

daytime

Cloud or snow? Ice or water?

Thermal infrared

day and night

How cold, so how high?

Every SEVIRI image has 12 channels: 11 with 3 km pixels (about 3×5 km over France) and the visible HRV with 1 km pixels (about 1×2 km). MTG has 16.

What can be downloaded

	Every	Years	Whole archive
Images: what the instrument measures			
SEVIRI rapid scan	5 min	May 2008 →	101 TB
SEVIRI, copy by Open Climate Fix	5 min	2008–2022	62 TB
MTG	10 min	Sep 2024 →	107 TB
Products: one answer per pixel, by EUMETSAT			
SEVIRI cloud mask	5 min	Oct 2008 →	0.15 TB
SEVIRI cloud-top height	15 min	Jun 2005 →	0.24 TB
MTG cloud mask	10 min	Jan 2025 →	0.45 TB
MTG cloud-top height, cloud type	10 min	Dec 2025 →	3.9 TB

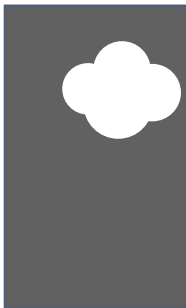
First product in the EUMETSAT Data Store (6 Oct 2026); MTG operational since 4 Dec 2024.

Two ways to get the same SEVIRI images

	EUMETSAT	Open Climate Fix copy
Years	2008 → today	2008–2022
Archive 2008–2022	≈ 81 TB	≈ 62 TB
One download	all Europe, one scan	1 hour, 100×100 pixels
Our cells, one day	≈ 8.4 GB	≈ 0.09 GB (measured)

Similar total size; the difference is the cut: the copy lets us download only the blocks over our cells.

One pixel, two answers: image or cloud mask



a 3 × 5 km pixel,
15% cumulus

Image $0.85 \times 10\% + 0.15 \times 70\% = \mathbf{19\%}$

Brighter than clear ground (10%): the cloud is still there.

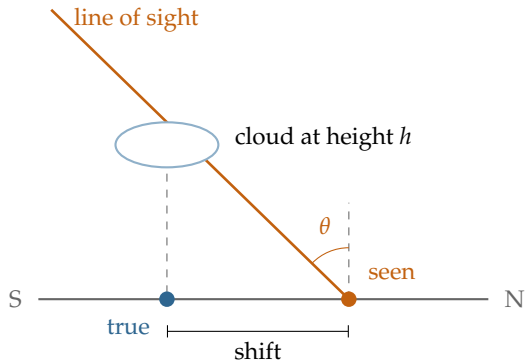
Cloud mask cloud or clear, nothing in between

Mostly “cloud” for all 15 km²; “clear” for the smallest cumuli.

We want the images, the HRV above all: the mask throws away how much cloud there is, and where.

Illustrative reflectances. Mask behaviour: Bley and Deneke (2013).

Parallax: clouds appear north of where they are



$$\text{shift} = h \tan \theta \approx 1.2\text{--}1.5 \times h$$

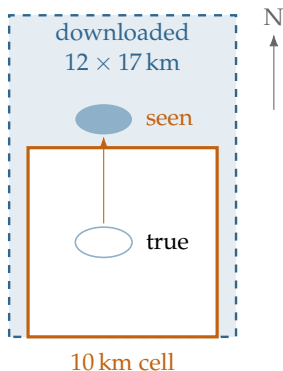
θ : angle of the line of sight from vertical,
50–57° over France

Cloud at 2.5 km 3–4 km north

Cloud at 4 km 5–6 km north

A cloud above a 10 km cell can show up in the next cell to the north.

So we download a larger window and shift it back

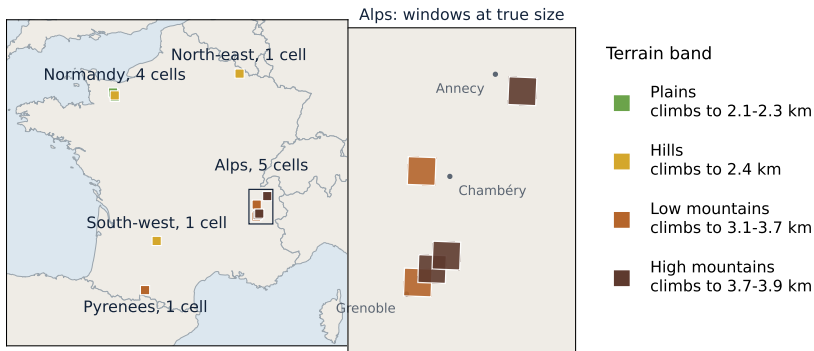


- 1 Download the cell plus 7 km to the north and 1 km on each side
- 2 Take the cloud height from the climbs of that day: satellite height maps miss small cumuli
- 3 Shift each image back by $1.2-1.5 \times h$, in Python with satpy

The margin costs a little more data; the correction is one function call, and can change later without downloading again.

satpy.modifiers.parallax: `get_parallax_corrected_lonlats` takes the satellite position, the pixel positions and a cloud height.

Our study: twelve 10 km windows



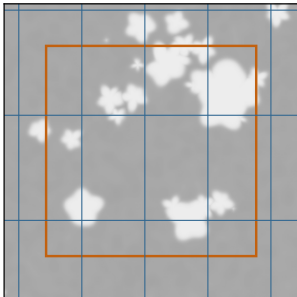
Every 5 min? 15min?, 2008–2026? 2008–2023?, 08:00–20:00 ?

Terrain band: highest IGN DEM elevation inside each 5 km cell.

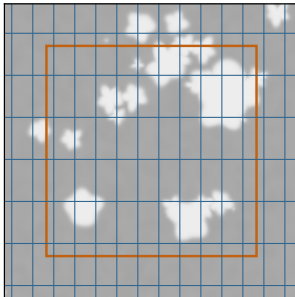
How the pixels fall on a 10 km cell

Pixel grids over a 10 x 10 km cell (orange); background: synthetic clouds

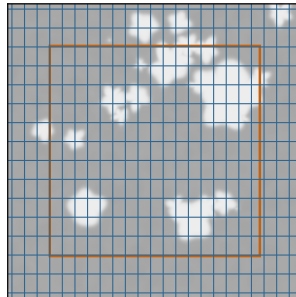
SEVIRI VIS0.6, 3 x 5 km
12 pixels touch the cell



SEVIRI HRV, 1 x 2 km
66 pixels touch the cell



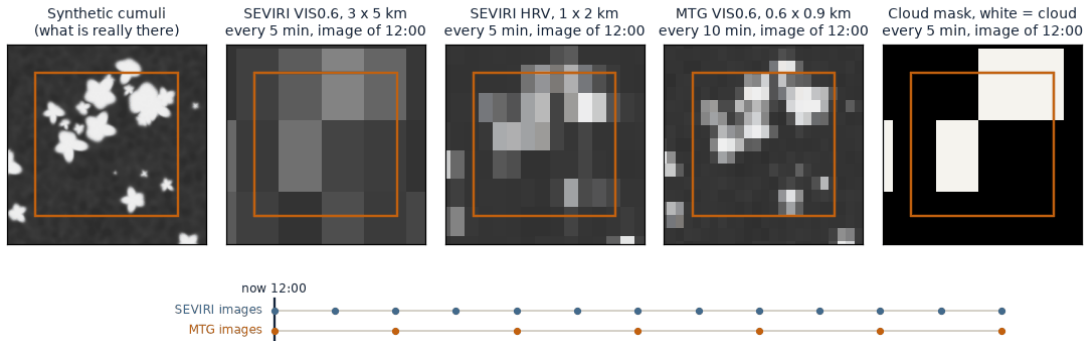
MTG VIS0.6, 0.6 x 0.9 km
221 pixels touch the cell



Cumuli are 0.5–2 km wide: SEVIRI's 3 km channels are too coarse for our analyses; its HRV channel sees the larger cumuli, MTG single clouds.

What each dataset would show in a 10 km cell, over one hour

Illustration of what to expect: synthetic clouds, not real satellite data



[Open GIF](#)

Illustration, not data. At 3×5 km, SEVIRI is too coarse for our analyses; HRV shows the larger cumuli, MTG single clouds. Each panel changes only when its satellite takes an image.

Three quarters of our flight-days fall in the cheap years



2008–2022

SEVIRI copy: cheap

2023–2024

SEVIRI files: expensive

2025–2026

MTG: feasible

SEVIRI runs to today; only its cheap copy stops in 2022. The cloud mask covers 2023–2024 cheaply.

	Kept	Downloaded
1 SEVIRI images 2008–2022, OCF copy	5.4 GB	420 GB
2 SEVIRI cloud mask 2008–2026	0.16 GB	85 GB
3 MTG images 2024–2026, 3 strips	1.1 GB	3.3 TB
4 Optional: SEVIRI cloud-top height	0.13 GB	149 GB
Total	≈ 7 GB	≈ 4.0 TB

Left out: SEVIRI images for 2023–2026, 11.6 TB to download for 1.2 GB kept.

EUMETSAT Data Store: product descriptions, counts and file sizes	queried 6 Oct 2026
Open Climate Fix SEVIRI rapid-scan Zarr, public bucket	block sizes measured 6 Oct 2026
Open Climate Fix Satip: channel scaling constants	satip/constants.py
MTG FCI Level 1 Product User Guide	40 strips per scan
Meteosat-12 declared operational	4 Dec 2024
MTG-I2 launch	27 Aug 2026
Bley and Deneke (2013), HRV cloud mask	EUMETSAT mask and sub-pixel cumuli
WMO OSCAR: SEVIRI	sampling and field of view
satpy parallax correction	needs cloud-top height

Cells and flight-days: viewer store thermal-planes.sqlite3. Pixel sizes, parallax shifts and download times are computed, except where marked as measured. Cloud drawings and animation are synthetic.